φ-NTT: A Carry-Free Transform on $\mathbb{Z}_{10}^B$ with Hierarchical Wavelet Structure

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Abstract

We introduce φ-NTT (phi Number Theoretic Transform), a transform defined over the carry-free direct product group $\mathbb{Z}_{10}^B$ with coefficients in the golden-ratio integer ring $\mathbb{Z}[\varphi] = \mathbb{Z}[\sqrt{5}]$. The transform achieves exact integer arithmetic with no floating-point error and no inter-stage twiddle factors. The B-stage construction yields a 2^B channel filterbank with a hierarchical wavelet structure analogous to Haar MRA. We establish a *Spectral Concentration Quasi-Theorem*: for signals with dominant period $P \approx 2 \cdot 10^b$, the detail channel at level b captures energy fraction $\alpha_b \geq C(1-\varepsilon)^{B-1}$. Empirically, the top 3 channels cover $\geq 97\%$ of total energy for $B = 3, 4, 5$.

1 Introduction

The Fast Fourier Transform (FFT) and its variants are among the most widely deployed algorithms in signal processing and communications. Standard FFT operates on cyclic groups $\mathbb{Z}/(N)$ and relies on complex-valued twiddle factors, which introduce floating-point error and prevent exact integer arithmetic.

In this paper, we introduce the *φ-NTT* (phi Number Theoretic Transform), a transform defined over the direct product group $\mathbb{Z}_{10}^B = \mathbb{Z}_{10} \times \cdots \times \mathbb{Z}_{10}$ (the group of B-digit decimal representations with carry-free, digit-wise modular arithmetic). The key structural features of φ-NTT are:

- **Exact integer arithmetic.** All coefficients lie in the ring $\mathbb{Z}[\varphi] = \mathbb{Z}[\sqrt{5}]$ (golden-ratio integer ring); no floating-point error occurs.
- **Zero inter-stage twiddle factors.** The tensor-product theorem eliminates all cross-stage phase corrections across all B recursive stages.
- **2^B channel filterbank.** The B-stage transform decomposes a length-10^B signal into 2^B channels, labeled by binary strings over $\{T, U\}^B$, where T denotes a C_5-twisted low-pass projection and U denotes a $\mathbb{Z}_2$ high-pass (parity) projection.

The carry-free arithmetic setting—where each decimal digit evolves independently—naturally gives rise to a hierarchical wavelet structure analogous to Haar multiresolution analysis (MRA), but defined over the ring $\mathbb{Z}[\varphi]$ and indexed by decimal digits rather than binary positions. We formalize this analogy and prove a Spectral Concentration Quasi-Theorem: for signals with dominant period $P \approx 2 \cdot 10^b$, the detail channel at level b captures a fraction $\alpha_b \geq C(1-\varepsilon)^{B-1}$ of total signal energy. Empirically, the top 3 channels cover $\geq 97\%$ of energy for $B = 3, 4, 5$.

The remainder of this paper is organized as follows. Section **??** defines φ-NTT and the convolution theorem on $\mathbb{Z}_{10}^B$. Section **??** develops the hierarchical wavelet interpretation. Section **??**

states and sketches the proof of the quasi-theorem. Section **??** presents numerical experiments for $B = 1, \ldots, 5$. Section **??** discusses open problems, including the exact orthogonality of T/U projections and the determination of constants C and ε. Section **??** concludes.

2 φ-NTT on $\mathbb{Z}_{10}^B$

2.1 Carry-Free Group Structure

Let B be a positive integer and $N = 10^B$. We represent each index $n \in \{0, \ldots, N-1\}$ by its decimal digit expansion

$$n = d_0(n) + 10\, d_1(n) + \cdots + 10^{B-1}\, d_{B-1}(n), \qquad d_i(n) \in \{0, \ldots, 9\}. \tag{1}$$

The *carry-free group* $\mathbb{Z}_{10}^B$ is the direct product $\mathbb{Z}_{10} \times \cdots \times \mathbb{Z}_{10}$ (B copies), with addition defined digit-wise:

$$(n \oplus m)_i = \big(d_i(n) + d_i(m)\big) \bmod 10. \tag{2}$$

This differs fundamentally from ordinary modular arithmetic on $\mathbb{Z}/(10^B)\mathbb{Z}$, where addition carries across digit boundaries. The two groups coincide only at $B = 1$; for $B \geq 2$ they are non-isomorphic, and the standard cyclic convolution theorem does not apply to $\mathbb{Z}_{10}^B$.

Carry-free subtraction and negation are defined analogously:

$$\text{cf_sub}(n, m)_i = \big(d_i(n) - d_i(m)\big) \bmod 10, \tag{3}$$

$$\text{cf_neg}(n)_i = \big(-d_i(n)\big) \bmod 10. \tag{4}$$

All arithmetic in φ-NTT operates over the ring $\mathbb{Z}[\varphi] = \mathbb{Z}[\sqrt{5}]$, where $\varphi = (1 + \sqrt{5})/2$. Elements are pairs $(a, b) \in \mathbb{Z}^2$ representing $a + b\varphi$, with multiplication

$$(a, b) \cdot (c, d) = (ac + bd,\ ad + bc + bd). \tag{5}$$

This representation eliminates floating-point error entirely.

2.2 Definition of φ-NTT

For $B = 1$ (length $N = 10$), the transform decomposes the input $x : \mathbb{Z}_{10} \to \mathbb{Z}[\varphi]$ into two channels via the isomorphism $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ (Chinese Remainder Theorem):

$$T\text{-channel:} \quad X_T[k] = \text{n10_T}(x)[k] \quad (C_5\text{-twisted DFT projection}), \tag{6}$$

$$U\text{-channel:} \quad X_U[k] = \text{n10_U}(x)[k] \quad (\mathbb{Z}_2 \text{ parity projection}). \tag{7}$$

The C_5-twisted DFT uses the basis elements $\text{cos5_zphi}(m) \in \{(2, 0), (-1, 1), (0, -1)\}$—three distinct values in $\mathbb{Z}[\varphi]$, with no irrational twiddle factors at runtime. The $\mathbb{Z}_2$ projection uses the parity kernel $U_1[k] \in \{(0, 0), (\pm 1, 0), (0, \pm 1)\}$. Explicit tables are given in Appendix **??**.

For general B, the transform is defined by a B-stage tensor product. At each stage $b \in \{0, \ldots, B-1\}$, the input is projected onto either the T-channel (low-pass, C_5 side) or the U-channel (high-pass, $\mathbb{Z}_2$ side), yielding 2^B output channels labeled by $s \in \{T, U\}^B$:

$$\text{bn_forward_flat}(x, B) \longrightarrow \big[\, \text{ch}_s : s \in \{T, U\}^B \,\big], \quad \text{ch}_s : \mathbb{Z}_{10}^B \to \mathbb{Z}[\varphi]. \tag{8}$$

The inverse transform bn_inverse_flat reconstructs x exactly, with scale factor 40^B (no convolution) or $80^B = 40^B \times 2^B$ (after channel-domain convolution). Both scale factors divide exactly in $\mathbb{Z}[\varphi]$, so reconstruction is lossless over the integers.

A key structural property is that *no inter-stage twiddle factors appear*: the tensor-product theorem over $\mathbb{Z}_{10}^B$ guarantees that all cross-stage phase corrections vanish across all B recursive stages.

2.3 Convolution Theorem (Carry-Free)

Define the carry-free convolution of $x, h : \mathbb{Z}_{10}^B \to \mathbb{Z}[\varphi]$ as

$$(x \star h)[n] = \sum_m h[m] \cdot x\big[\mathrm{cf_sub}(n, m)\big], \qquad n \in \mathbb{Z}_{10}^B. \tag{9}$$

Theorem 2.1 (Carry-Free Convolution)**.** *Let F_B denote the φ-NTT forward transform. Then F_B diagonalizes carry-free convolution on $\mathbb{Z}_{10}^B$: the channel-domain product*

$$Y_s[k] = X_s[k] \cdot H_s[k] \quad \textit{(with the } T/U \textit{ cross-coupling rule)}$$

recovers $x \star h$ upon inverse transform, up to scale factor 80^B. The T/U cross-coupling at each stage is:

$$2\,Y_T[k] = X_T[k]\,H_T[k] - \mathrm{SIN2} \cdot X_U[k]\,H_U[k], \tag{10}$$

$$2\,Y_U[k] = X_T[k]\,H_U[k] + X_U[k]\,H_T[k], \tag{11}$$

where $\mathrm{SIN2} = (3, -1) \in \mathbb{Z}[\varphi] \approx 1.382 = 4\sin^2(2\pi/10)$.

Theorem **??** has been verified for $B = 1, \ldots, 5$ against direct computation. Similarly, carry-free cross-correlation reduces to convolution with the carry-free-reversed kernel $h_{\mathrm{rev}}[m] = h[\mathrm{cf_neg}(m)]$.

3 Hierarchical Wavelet Interpretation

The recursive T/U decomposition induces a hierarchical structure on $\mathbb{Z}_{10}^B$ that parallels multiresolution analysis (MRA).

3.1 T/U as Projection-Like Operators

At each stage $b \in \{0, \ldots, B-1\}$, the φ-NTT applies one of two projection-like operators to digit d_b:

$$\begin{aligned} &T_b : \text{low-pass side} \quad (\text{extracts the slowly-varying component across } d_b \in \{0, \ldots, 9\}),\\ &U_b : \text{high-pass side} \quad (\text{extracts the alternating component } (-1)^{d_b}). \end{aligned}$$

Together, T_b and U_b partition the information in digit b in a manner analogous to a two-channel filter bank. We use the term *projection-like* deliberately: while numerical experiments confirm that energy is partitioned across channels without loss (exact reconstruction holds for all $B = 1, \ldots, 5$), the strict algebraic orthogonality $T_b \perp U_b$ has not yet been proved and is left as an open problem (Section **??**).

The B-stage tensor product applies these operators independently at each digit position, yielding 2^B channels:

$$\mathrm{ch}_s = \Big(\bigotimes_{b=0}^{B-1} P_b^{s_b}\Big)(x), \qquad s \in \{T, U\}^B, \tag{12}$$

where $P_b^T = T_b$ and $P_b^U = U_b$.

3.2 Binary Tree Interpretation

The label set $\{T, U\}^B$ is in natural bijection with the set of root-to-leaf paths in a complete binary tree of depth B. Reading the label from left (most significant digit, $b = B-1$) to right (least significant, $b = 0$), the tree has the form:

```
        root
      /       \
T(d_{B-1})  U(d_{B-1})
  /   \       /   \
 T     U    T     U
(d_{B-2})  ...
```

The channel $TT\cdots T$ (all T, leftmost leaf) captures the coarsest approximation; the channel $UU\cdots U$ (all U, rightmost leaf) captures the finest detail—the checkerboard pattern across all digit positions simultaneously.

Table **??** contrasts the φ-NTT with Haar MRA.

Table 1: Comparison of Haar MRA and φ-NTT.

Property	Haar MRA	φ-NTT
Signal length	2^B	10^B
Channels	2^B	2^B
Approximation coeff.	scaling function	$TT\cdots T$ channel
Detail coeff.	wavelet function	$T\cdots TU_bT\cdots T$ channels
Arithmetic	real (± 1 basis)	$\mathbb{Z}[\varphi]$ basis
Orthogonality	exact	conjectured

3.3 Energy Localization Mechanism

Smooth signals vary slowly across all digit positions. The U_b operator measures the parity alternation $(-1)^{d_b}$ at digit b; if x changes little between even and odd values of d_b, the U_b inner product is small. Accumulating this across B stages, the $UU\cdots U$ channel is driven toward zero, explaining the empirical observation (B = 3, 4, 5) that $UU\cdots U$ carries $< 0.01\%$ of total energy for smooth signals.

Signals with a dominant period $P \approx 2\cdot 10^b$ exhibit strong alternation specifically at digit b, while other digits vary slowly. The U_b projection then captures the bulk of the oscillatory energy, while T projections at all other stages pass the slowly varying envelope. The result is that the channel $T\cdots T[U_b]T\cdots T$ (U only at position b) becomes dominant. For the test signal $\sin(2\pi n/100)$ with $P = 100 \approx 2\cdot 10^1$ ($b = 1$), this predicts the $TUT\cdots T$ channel as dominant—consistent with all $B = 3, 4, 5$ experiments (86%, 69%, 68% of total energy, respectively).

4 Spectral Concentration Quasi-Theorem

We formalize the observed concentration phenomenon.

4.1 Statement

We work under two assumptions on the input signal $x : \mathbb{Z}_{10}^B \to \mathbb{Z}[\varphi]$ (real-valued, i.e., second component identically 0):

(A1) **Digit-wise smoothness.** For each digit position $b' \neq b$, the variation of x across $d_{b'}$ is small relative to the variation across d_b:

$$|\text{variation}(x, b')| \leq \varepsilon \cdot |\text{variation}(x, b)|, \qquad b' \neq b,$$

for some constant $0 < \varepsilon < 1$.

(A2) **Dominant period.** The signal has a dominant period $P \approx 2 \cdot 10^b$ for some $b \in \{0, \ldots, B-1\}$, so that d_b is the primary source of oscillation.

Theorem 4.1 (φ-NTT Spectral Concentration, Quasi-Theorem)**.** *Under assumptions* (A1) *and* (A2)*, there exist constants $C > 0$ and $0 < \varepsilon < 1$ (depending on the signal class) such that the energy fraction captured by the level-b detail channel satisfies*

$$\alpha_b \;=\; \frac{E\big[\mathrm{ch}_{T\cdots TU_bT\cdots T}\big]}{E_{\mathrm{total}}} \;\geq\; C \cdot (1-\varepsilon)^{B-1}. \tag{13}$$

Furthermore, the top 3 channels (by energy) jointly satisfy

$$\alpha_b + \alpha_{b,2} + \alpha_{b,3} \;\geq\; 1 - \delta \tag{14}$$

for a small residual δ, empirically $\delta < 0.03$ for $B = 3, 4, 5$.

We call this a *quasi-theorem* because the constants C and ε are characterized implicitly through (A1)–(A2) rather than in closed form, and because the strict algebraic orthogonality of the T/U decomposition—while consistent with all numerical evidence—remains unproved (see Section **??**).

4.2 Interpretation

The bound $C(1-\varepsilon)^{B-1}$ has a transparent structure: ε measures digit-level roughness—how much energy leaks from the dominant digit b to each of the other $B-1$ digits. The factor $(1-\varepsilon)^{B-1}$ is the product of $B-1$ per-stage suppression factors, explaining why the observed α_b decreases from 86% ($B = 3$) to 68% ($B = 4, 5$) as more stages accumulate leakage. The constant C absorbs the baseline energy ratio at the dominant stage itself.

The exponential decay in B is a structural feature: with more digit stages, more channels compete for energy. The top-3 coverage ($\geq 97\%$) remains stable because the channels at levels $b \pm 1$ absorb most of the leakage.

4.3 Proof Sketch

We sketch the structure of the argument; a complete algebraic proof is left for future work.

Step 1 (Digit smoothness $\Rightarrow$ U energy bound). By (A1), for each $b' \neq b$, the $U_{b'}$ projection has small inner product with x, giving $E[U_{b'}] \leq \varepsilon^2 \cdot E[U_b]$ for $b' \neq b$.

Step 2 (Dominant digit $\Rightarrow$ U_b lower bound). By (A2), x oscillates with period $\approx 2 \cdot 10^b$, correlating strongly with the $\mathbb{Z}_2$ parity indicator $(-1)^{d_b}$. This is precisely the kernel of U_b, yielding $E[U_b] \geq C_0 \cdot E_{\mathrm{total}}$ for some $C_0 > 0$.

Step 3 (Tensor product $\Rightarrow$ multiplicative separation). Because the B-stage decomposition applies each operator independently per digit, the channel $T \cdots TU_bT \cdots T$ receives energy from U_b at stage b and T at all other stages. The T projections at stages $b' \neq b$ pass through the slowly varying envelope (factor ≈ 1), while $U_{b'}$ channels at those stages are suppressed by ε per stage.

Step 4 (Normalization $\Rightarrow$ α_b lower bound). Summing the energy across all 2^B channels and normalizing yields $\alpha_b \geq C(1-\varepsilon)^{B-1}$, where C absorbs C_0 and the T-stage pass-through factors.

4.4 Empirical Alignment

The quasi-theorem is consistent with all numerical experiments ($B = 3, 4, 5$; test signal $\sin(2\pi n/100)$, $P = 100 \approx 2 \cdot 10^1$, $b = 1$):

In all cases, the top 3 channels jointly cover $\geq 97\%$ of total energy, and the $UU \cdots U$ channel carries $< 0.01\%$ in all experiments. The values of C and ε are not fitted here; the table shows only that the observed decay is consistent with the bound structure $C(1-\varepsilon)^{B-1}$.

Table 2: Empirical alignment with Quasi-Theorem **??**.

B	α_b (observed)	Lower bound structure
3	0.864	$C(1-\varepsilon)^2$ with $\varepsilon \approx 0.07$
4	0.687	$C(1-\varepsilon)^3$
5	0.682	$C(1-\varepsilon)^4$

5 Numerical Experiments

5.1 Experimental Setup

All experiments use the reference implementation `phi_carry_free.py`, operating entirely in exact integer arithmetic over $\mathbb{Z}[\varphi]$. No floating-point operations are involved in the transform itself; `zval()` (evaluation to float) is used only for energy measurement in post-processing.

We test $B = 1, \ldots, 5$ (signal lengths $N = 10$ to 100,000) with a low-frequency test signal

$$x[n] = \sin(2\pi n/100) + \sin(2\pi n/P_2) + \text{noise},$$

where P_2 is a secondary period scaled to N and the noise is a small integer perturbation. This signal class satisfies assumptions (A1)–(A2) of Section **??** with dominant period $P = 100 \approx 2 \cdot 10^1$ ($b = 1$).

Correctness is verified in two ways: (i) direct comparison of `phi_conv_carry_free` against the reference $O(N^2)$ carry-free convolution, and (ii) round-trip test with a delta kernel $h = \delta_0$, confirming exact reconstruction for all $B = 1, \ldots, 5$.

5.2 Channel Energy Distribution

Table **??** shows the energy fraction of the top-3 channels and the $UU \cdots U$ channel for $B = 3, 4, 5$. All fractions are computed as

$$E[\text{ch}]/E_{\text{total}}, \qquad E[\text{ch}] = \frac{1}{N} \sum_k \big(\text{zval}(\text{ch}[k])\big)^2.$$

Table 3: Channel energy distribution (low-frequency test signal).

B	#1 ch	α_1	#2 ch	α_2	#3 ch	α_3	Top-3	$UU\cdots U$
3	TUT	86.4%	TTT	11.1%	UTT	2.3%	99.8%	$< 0.01\%$
4	TUTT	68.7%	TTUT	17.7%	TTTT	11.7%	98.1%	$< 0.01\%$
5	TUTTT	68.2%	TTTUT	17.7%	TTTTT	11.7%	97.6%	$< 0.01\%$

Three observations are immediate. First, the dominant channel is consistently of the form $T \cdots TU_1T \cdots T$ (U only at digit $b = 1$), in agreement with the Section **??** prediction for $P \approx 2 \cdot 10^1$. Second, the top-3 channels jointly cover $\geq 97\%$ of total energy across all tested values of B, consistent with the $\delta < 0.03$ claim of Theorem **??**. Third, the $UU \cdots U$ channel carries negligible energy ($< 0.01\%$) in all cases, consistent with the checkerboard suppression argument of Section **??**.

The "∞ (exact)" entries confirm lossless reconstruction when all channels are retained.

5.3 Reconstruction Accuracy

Exact reconstruction is guaranteed by the algebraic structure of φ-NTT, not by numerical precision. The scale factors 40^B (round-trip) and 80^B (after convolution) divide exactly in $\mathbb{Z}[\varphi]$, as verified by assertion checks for all $B = 1, \ldots, 5$:

Table 4: SNR vs. retained channels (low-frequency signal, $B = 3, 4, 5$).

B	1 ch	2 ch	4 ch	All ch
3	9.5 dB	15.4 dB	28.5 dB	∞ (exact)
4	5.3 dB	9.5 dB	20.5 dB	∞ (exact)
5	5.2 dB	9.3 dB	18.5 dB	∞ (exact)

```
assert v[0] % scale == 0 and v[1] % scale == 0   # holds for all v
```

The delta-kernel test `phi_conv_carry_free(x, `δ_0`, B) == x` passes for all tested inputs and all $B = 1, \ldots, 5$, confirming Theorem **??**.

5.4 Computational Structure

The φ-NTT avoids two sources of complexity common in FFT implementations. First, there are no inter-stage twiddle factors: all phase corrections vanish by the tensor-product theorem over $\mathbb{Z}_{10}^B$. Second, all arithmetic is integer-valued; the only divisions are the final scale-factor normalizations, which are exact. The transform operates on length-10^B signals with 2^B output channels in $O(N \cdot B)$ integer operations per stage. Detailed complexity analysis and comparison with FFT are reserved for future work.

6 Discussion

6.1 Open Problems

Three problems remain open.

(P1) Orthogonality of T/U projections. The numerical experiments confirm that energy is partitioned across channels without loss and that exact reconstruction holds for all tested B. However, the strict algebraic orthogonality $T_b \perp U_b$ over $\mathbb{Z}[\varphi]$ has not been proved. Establishing this would upgrade Theorem **??** to a full theorem, with C and ε determined by the projection geometry.

(P2) Closed-form determination of C and ε. The constants C and ε in Theorem **??** are currently characterized implicitly through (A1)–(A2). Deriving explicit expressions—presumably in terms of the signal period P, the digit base 10, and the $\mathbb{Z}[\varphi]$ basis elements—would make the spectral concentration bound fully quantitative and signal-class independent.

(P3) Connection to carry-based convolution (overlap-add). The present work is restricted to carry-free convolution on $\mathbb{Z}_{10}^B$. A natural extension is to bridge this to ordinary cyclic convolution on $\mathbb{Z}/(10^B)\mathbb{Z}$ via an overlap-add scheme, enabling φ-NTT-based filtering of standard integer sequences. The group non-isomorphism $\mathbb{Z}_{10}^B \ncong \mathbb{Z}/(10^B)\mathbb{Z}$ (for $B \geq 2$) means this bridge is non-trivial and requires a separate treatment.

6.2 Relation to Existing Frameworks

vs. FFT. Standard FFT operates on the cyclic group $\mathbb{Z}/(N)\mathbb{Z}$ with complex twiddle factors, accumulating floating-point error. φ-NTT operates on $\mathbb{Z}_{10}^B$ with coefficients in $\mathbb{Z}[\varphi]$, achieving exact integer arithmetic. The two transforms diagonalize different convolution operations and are not directly comparable in terms of computational cost; FFT runs in $O(N \log N)$ while the complexity of φ-NTT is $O(NB)$ with $N = 10^B$.

vs. Standard NTTs. Galois-field NTTs (e.g., used in lattice-based cryptography) achieve exact integer arithmetic by working modulo a prime p. φ-NTT works over $\mathbb{Z}[\varphi]$ without modular reduction, preserving the full digit structure of $\mathbb{Z}_{10}^B$. The two frameworks address different algebraic settings and are complementary.

vs. Haar Wavelets. Haar MRA on length-2^B signals uses ± 1 arithmetic and achieves exact orthogonality. φ-NTT on length-10^B signals uses $\mathbb{Z}[\varphi]$ arithmetic and conjectured quasi-orthogonality. The structural analogy (Section **??**) is precise; the arithmetic and group structure differ.

6.3 Limitations

The current work has the following limitations. The quasi-theorem (Theorem **??**) is stated for real-valued signals satisfying (A1)–(A2); extension to complex-valued or high-bandwidth signals remains open. The computational complexity $O(NB)$ has not been compared to FFT implementations for the same problem size. The carry-free convolution studied here does not directly apply to standard digital filtering tasks, which rely on ordinary cyclic convolution (Problem P3).

7 Conclusion

We have introduced φ-NTT, a transform defined over the carry-free direct product group $\mathbb{Z}_{10}^B$ with coefficients in the golden-ratio integer ring $\mathbb{Z}[\varphi]$. The three main contributions are as follows.

First, we establish a carry-free convolution theorem (Theorem **??**): φ-NTT diagonalizes carry-free convolution on $\mathbb{Z}_{10}^B$, with exact integer arithmetic and no floating-point error.

Second, we develop a hierarchical wavelet interpretation (Section **??**): the B-stage T/U decomposition is structurally analogous to Haar MRA, producing 2^B channels arranged in a complete binary tree of depth B.

Third, we state a Spectral Concentration Quasi-Theorem (Theorem **??**): for signals with dominant period $P \approx 2 \cdot 10^b$, the level-b detail channel captures energy fraction $\alpha_b \geq C(1-\varepsilon)^{B-1}$, and the top 3 channels jointly cover $\geq 97\%$ of total energy for $B = 3, 4, 5$.

Open problems include the algebraic proof of T/U orthogonality (P1), closed-form determination of C and ε (P2), and the connection to standard carry-based convolution via overlap-add (P3). We hope this work opens a productive line of inquiry at the intersection of carry-free arithmetic, wavelet theory, and exact integer transforms.

A Basis Tables for $B = 1$

A.1 C_5-Twisted DFT Basis (T-channel)

The T-channel basis elements $\text{cos5_zphi}(m) \in \mathbb{Z}[\varphi]$ for $m \in \{0, 1, 2, 3, 4\}$ (one representative per C_5 orbit) take exactly three distinct values:

$$\text{cos5_zphi}(m) \in \{(2, 0),\ (-1, 1),\ (0, -1)\} \subset \mathbb{Z}[\varphi],$$

corresponding to $2\cos(0) = 2$, $2\cos(2\pi/5) = \varphi - 1$, and $2\cos(4\pi/5) = -\varphi^{-1}$ respectively, expressed as elements of $\mathbb{Z}[\varphi]$. No irrational twiddle factors appear at runtime.

A.2 $\mathbb{Z}_2$ Parity Kernel (U-channel)

The U-channel parity kernel $U_1[k]$ for $k \in \{0, \ldots, 9\}$ takes values in $\{(0, 0), (\pm 1, 0), (0, \pm 1)\} \subset \mathbb{Z}[\varphi]$, corresponding to the parity indicator $(-1)^k$ lifted into $\mathbb{Z}[\varphi]$ via the even/odd split of $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$.

B Permutation `perm10`

The permutation `perm10` reindexes $\{0, \ldots, 9\}$ to align the CRT decomposition $\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5$ with the natural digit order. Explicitly, `perm10` maps $k \mapsto \sigma(k)$ where $\sigma = (0, 5, 2, 7, 4, 9, 6, 1, 8, 3)$ (0-indexed). This permutation is its own inverse up to sign, and all φ-NTT stages apply it before the C_5 DFT step. The full derivation is given in the reference implementation `phi_carry_free.py`.

C 80^B Scale Factor Decomposition

The scale factor 80^B arising from a full forward–inverse cycle with carry-free convolution decomposes as

$$80^B = 40^B \times 2^B,$$

where 40^B arises from the B-stage forward–inverse round trip (each stage contributing a factor of 40) and the additional 2^B arises from the T/U cross-coupling normalization at each of the B stages. Both factors are exact divisors in $\mathbb{Z}[\varphi]$ (no remainder), ensuring lossless integer reconstruction. The complete verification that 80^B divides all output elements exactly is implemented in the assertion checks of `phi_carry_free.py` and holds for all tested $B = 1, \ldots, 5$.